Weak-Memory Local Reasoning, Revisited

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Abstract

My 2012 dissertation proposed a program logic for local reasoning about concurrent programs on an x86-like, total-store-order memory model. It left two components unfinished: the soundness proof, and a proof system for assertions. This report returns to both with a mechanization in Rocq. The soundness theorem does not hold as printed: four of the printed rules derive false specifications. A repaired calculus is sound, with operational adequacy, and keeps the assertion language, the unit law for **emp**, the temporal frame rule and the printed load and store axioms. The report explains what was wrong, what had to change and why, and what the repaired logic can prove.

1 Introduction

The dissertation [1] set out to give a local reasoning principle for programs running on an x86-like machine, where each processor buffers its writes and other processors cannot see a write until it leaves the buffer. Its logic has three separating conjunctions. Spatial conjunction $P * Q$ splits memory by address, as in separation logic. Temporal conjunction $P \lhd Q$ splits it by time: P describes earlier writes and Q later ones, possibly to the same addresses. Strong temporal conjunction $P \blacktriangleleft Q$ requires both. Each has a frame rule, and a concurrency rule and a shared invariant in the style of concurrent separation logic sit on top.

The dissertation was candid about where it stopped. Its statement of contributions lists two major components as unfinished and admits a third gap:

1. there is no proof system for the assertion language;
2. the soundness proof for the specification logic is incomplete; and
3. the document "gives little indication of how proofs ought to proceed in the logic."

It also hypothesized that the logic was sound and that the remaining proofs were "mostly straightforward inductions."

This report returns to those three items, fourteen years later, with a mechanization in the Rocq prover. Its findings are as follows.

The soundness theorem does not hold as printed. Several auxiliary claims are false, and each has a local repair (Section 3.1). More seriously, the rules themselves are inconsistent with any adequate semantics: from SKIP, CONS, INV and FRAME-TM alone one can derive that a skip changes the contents of a memory cell (Section 3.2). No choice of safety predicate can repair that.

The cause is general. An unrestricted temporal frame rule, a right unit for temporal conjunction, the rule of consequence, and any command that gives away ownership of a cell cannot coexist (Section 3.3). The observation is independent of the memory model. It applies to any logic that combines a non-cancellative sequential conjunction with ownership transfer.

A repaired calculus is sound. The repair replaces the invariant rule by lock and unlock rules that transfer the invariant, restricts the temporal frame rule to specifications that tolerate overlapping history, and adjusts four side conditions (Section 4). The repaired calculus is proved sound by induction on derivations, with an adequacy theorem for the dissertation's machine (Section 7). The proofs are checked by the Rocq kernel and use no axioms.

Most of the design survives. The machine, the assertion language and its interpretation of atoms, the unit laws for **emp**, overlapping temporal history, the temporal and spatial frame rules, the concurrency rule, and the printed load and store axioms are all retained. So is the dissertation's assertion algebra, with one family of exceptions: every printed law that rebrackets or exchanges through strong temporal conjunction fails, and with it the strong temporal frame rule (Appendix B). [*pending:* algebra table in the appendix]

The logic can be used. Section 8 derives a shared container in which a thread acquires the lock, removes a cell, releases the lock, writes the cell privately, and later returns it, while another processor's write stays buffered throughout. [*pending:* the Section 4.1 example; the larger derivation; the assertion proof system]

The report assumes a reader who knows the dissertation and has forgotten its details. Section 2 recalls the original model and calculus in the original notation. Sections 3 and 4 say what was wrong and what changed. Sections 5 to 8 present the repaired model, calculus, results and derivations. Proofs are not reproduced; every claim cites the name of its Rocq theorem, and the appendix gives fuller statements.

2 The original model and calculus

This section recalls Chapter 4 of the dissertation, in its notation. Nothing here is new. Page and figure numbers refer to the dissertation.

2.1 The machine

A *memory system* is a triple $\mu = (h, B, K)$: a heap $h : \mathbb{L} \rightharpoonup \mathbb{V}$, an array of write buffers $B : \mathbb{P} \to (\mathbb{L} \times \mathbb{V})$ list with the oldest write first, and a set $K \subseteq \mathbb{P}$ of *blocked* processors. A state pairs a stack s with a memory system. The lock is not a separate component: it is free when $K = \emptyset$ and held by i when $K = \mathbb{P} \setminus \{i\}$.

Primitive commands carry the identifier of the processor that executes them. Their behaviour on processor i is as follows.

- $[e] := e'_i$ appends a write to $B(i)$. The address must be allocated, in the heap or in $B(i)$. A blocked processor may still store.
- $x := [e]_i$ requires $i \notin K$ and returns the newest write to that address in $B(i)$, or else the heap value.
- fence_i, lock_i and unlock_i require $B(i) = \varepsilon$. lock_i additionally requires $K = \emptyset$ and sets $K = \mathbb{P} \setminus \{i\}$; unlock_i reverses this.
- A silent step $\underset{\tau}{\to}$ moves the oldest write of some unblocked processor from its buffer to the heap.

Commands are built from primitives by sequencing $c\,;c'$, choice $c + c'$, iteration c^* and parallel composition $c \parallel c'$. There is no allocation. While a processor holds the lock, no other processor can load or flush, but others may continue to store and to assign to registers.

2.2 Generalized memories and separation

A *generalized memory system* $\nu = (h, B, K, \Gamma)$ adds a *buffer-completeness* set $\Gamma \subseteq \mathbb{P}$, with $h = \emptyset$ whenever $\Gamma = \emptyset$. Membership $i \in \Gamma$ means that buffer i is complete: nothing may be placed before it.

Three operations combine generalized memories. Write $\nu_k = (h_k, B_k, K_k, \Gamma_k)$. All three take $K_0 \cup K_1$ and $\Gamma_0 \cup \Gamma_1$ and require $K_0 \cap K_1 = \emptyset$.

Spatial, $\nu_0 \mathbin{\widehat{*}} \nu_1$. The address domains of the two sides, counting both heaps and buffers, are disjoint. The heap is $h_0 \uplus h_1$ and each buffer is an interleaving of the two.

Temporal, $\nu_0 \mathbin{\widehat{\lhd}} \nu_1$. The heap is $h_0 \backslash\backslash h_1$, so later values win, and each buffer is $B_0(i) +\!\!+ B_1(i)$. The sides may share addresses, subject to a compatibility condition: an unblocked pending write in ν_0 may not target an address in the heap of ν_1, nor an address pending on a different unblocked processor in ν_1. Finally $B_0(i) = \varepsilon$ for every $i \in \Gamma_1$.

Spatiotemporal, $\nu_0 \mathbin{\widehat{\#}} \nu_1$. As temporal, but each buffer is any interleaving whose lookup function agrees with $B_0(i) +\!\!+ B_1(i)$, and the condition on Γ_1 is dropped. It contains both of the others and is used only in the semantics of specifications.

Strong temporal separation $\nu_0 \mathbin{\widehat{\blacktriangleleft}} \nu_1$ is $\nu_0 \mathbin{\widehat{\lhd}} \nu_1$ restricted to pairs that are also spatially compatible.

2.3 Assertions

Assertions are given by

$$\begin{aligned} P \ ::= \ & b \mid P \vee P \mid P \wedge P \mid \exists x : P \mid \forall x : P \mid \mathbf{emp} \mid \mathbf{bar}_e \mid \mathbf{lock}_e \mid e \rightsquigarrow_{e'} e'' \\ & \mid P * P \mid P \lhd P \mid P \mathrel{-\!\!*} P \mid P \mathrel{-\!\!\lhd} P \mid P \mathrel{\lhd\!\!-} P \end{aligned}$$

and are interpreted over a stack and a generalized memory. The atoms are:

- **emp**: empty heap, empty buffers, $K = \emptyset$; Γ is unconstrained.
- $\mathbf{bar}_i$: as **emp**, with $i \in \Gamma$ when i is a processor.
- $\mathbf{lock}_i$: empty heap and buffers, with $K = \mathbb{P} \setminus \{i\}$.
- $e \rightsquigarrow_i e'$ ("leads to"): either a single *pending* write (e, e') in $B(i)$ and nothing else, or a single *flushed* heap cell $e \mapsto e'$ with $i \in \Gamma$.

$P * Q$ and $P \lhd Q$ hold of ν when ν is a spatial, respectively temporal, combination of a model of P and a model of Q. The three implications are the corresponding adjoints. Two abbreviations matter below:

$$P \blacktriangleleft Q =_{df} (P * Q) \wedge (P \lhd Q) \qquad\qquad e \mapsto e' =_{df} (\exists x : e \rightsquigarrow_x e') \lhd (\forall x : \mathbf{bar}_x).$$

Points-to therefore describes a flushed cell with *every* processor buffer-complete. The dissertation claims that every assertion is closed under flushing (Proposition 9), and that each conjunction is associative with **emp** as a unit (Figure 4.5).

2.4 Specifications and rules

A specification $J \vdash \{P\}\ c\ \{Q\}$ has an invariant J for shared state and a private precondition and postcondition. Figure 1 reproduces the axioms and rules.

Three features of the design are worth recalling, because the repair turns on them. First, LOCK and UNLOCK produce and consume only the token $\mathbf{lock}_i$. The invariant reaches the lock holder through INV, which may be applied to any command executed while the lock is held and allows the private state to exchange cells with J. Second, FRAME-TM has only the usual variable condition: any history R may be placed before any specification. Third, temporal conjunction is not cancellative, as the dissertation remarks: $\mu \mathbin{\widetilde{\lhd}} \mu_0 = \mu \mathbin{\widetilde{\lhd}} \mu_1$ does not imply $\mu_0 = \mu_1$.

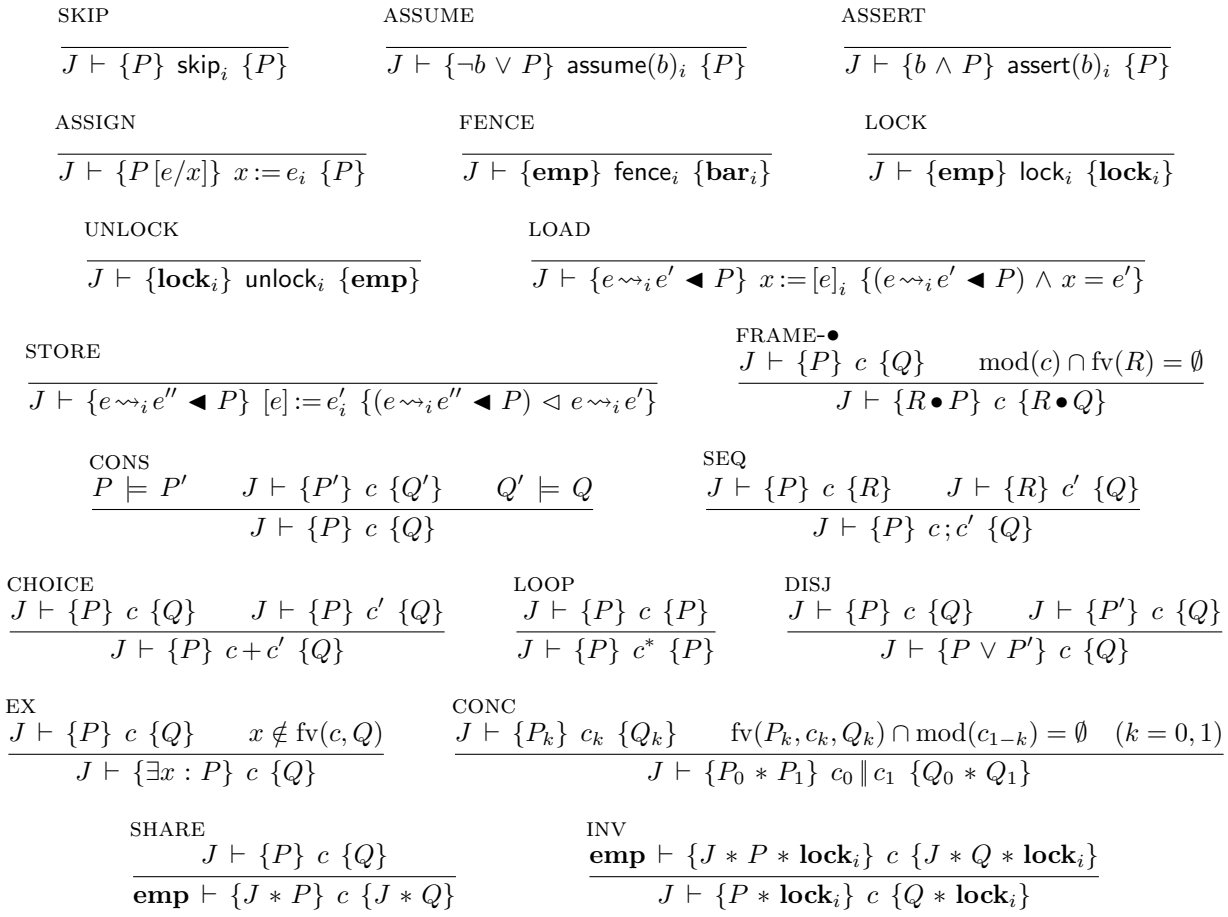


Figure 1: The printed axioms and rules (Figures 4.7 and 4.8). FRAME-• stands for three rules, FRAME-SP, FRAME-TM and FRAME-STM, with • read as $*$, $\lhd$ and $\blacktriangleleft$.

2.5 Semantics and status

Truth of a specification is defined through a family of predicates $safe_n(c, s, \mu, J, Q)$ after Vafeiadis. The predicate holds at $n+1$ when: if c is skip then $(s, \mu, \mathbb{P}) \models Q$; and for every μ_J, μ_F and $\mu_0 \in \mu_J \mathbin{\widetilde{*}} (\mu_F \mathbin{\widetilde{\#}} \mu)$ that is lock-complete, with either $(s, \mu_J, \mathbb{P}) \models J$ or the lock held in μ_0, the configuration does not abort, and every step to (c', s', μ_1) admits a new decomposition $\mu_1 \in \mu'_J \mathbin{\widetilde{*}} (\mu'_F \mathbin{\widetilde{\#}} \mu')$ with $\mu'_F \preceq \mu_F$, the same condition on J, and $safe_n(c', s', \mu', J, Q)$. A specification is true, $J \models \{P\}\ c\ \{Q\}$, when $(s, \mu, \mathbb{P}) \models P$ implies $safe_n(c, s, \mu, J, Q)$ for all n. Local states are always read at full completeness, $\Gamma = \mathbb{P}$.

Theorem 1 states that provable specifications are true. The dissertation gives soundness lemmas for the axioms in an appendix, sketches a frame lemma for the spatiotemporal operator $\widetilde{\#}$, and reduces the spatial frame rule to Conjecture 1, which is left open. Associativity of buffer overriding (Proposition 2) and of the three separations, and flushing closure of assertions (Proposition 9), are stated with at most a sketch.

3 What was wrong

The problems are presented from the most local to the most structural, which is also the order in which the mechanization found them. Each is a theorem checked by Rocq against a transcription of the printed definitions; the theorem's name is given so that the statement can be inspected.

3.1 Local errors with local repairs

Table 1 lists six printed claims that are false as stated. None is deep, and each has a repair that leaves the design intact.

Printed claim	Counterexample	Repair
ASSIGN is sound for every invariant	With J the assertion $x = 0$, the axiom derives $J \vdash \{\mathbf{true}\}\ x := 1_1\ \{\mathbf{true}\}$, and the assignment falsifies J with the lock free. `printed_assignment_invalid`	Require $x \notin \mathrm{fv}(J)$.
Buffer overriding is associative (Prop. 2)	Three one-element buffers writing 1, 2, 3 to one address: the list $[2,1,3]$ is in $a \mathbin{\backslash\!\backslash} (b \mathbin{\backslash\!\backslash} c)$ but not in $(a \mathbin{\backslash\!\backslash} b) \mathbin{\backslash\!\backslash} c$. `printed_buffer_override_not_associative`	A write from the later buffer may be taken only after the earlier buffer's writes to that address.
Spatiotemporal separation is associative	Compatibility ignores blocked processors, so whether two memories compose can depend on a third that supplies the blocking fact. `printed_spatiotemporal_not_associative`	Check conflicts whether or not the processor is blocked.
Assertions are closed under flushing (Prop. 9)	$\mathbf{false} \mathrel{\triangleleft\!\!-} \mathbf{bar}_1$ holds vacuously while processor 1 has a pending write, and fails once the write flushes. `pointwise_temporal_wand_not_flush_closed`	Implications quantify over all futures of the current memory.
Conjecture 1	A cell duplicated across $\widetilde{\#}$ with equal values is not a spatial split. `printed_conjecture1_false`	Not needed by the final proof.
UNLOCK is sound for every invariant	With $J = \mathbf{false}$ and the lock held, unlock_i is enabled and nothing restores J. `arbitrary_invariant_unlock_invalid`	See Section 4.

Table 1: Printed claims that fail, each with a checked counterexample.

Two remarks. The repaired compatibility relation is stronger than the printed one, so it rejects some compositions the dissertation admits. Those are states in which a blocked processor has a pending write to a cell that someone else owns, which arise only under the "daring" rules that the dissertation proposed and did not adopt. And the repaired implications are the ordinary Kripke reading of implication over the flushing preorder; the three adjunctions between conjunctions and implications still hold.

The failure of UNLOCK points at something larger. The printed safety predicate asks for the invariant only while the lock is free. Once a thread holds the lock, the predicate has forgotten that J held at the moment of acquisition, so neither UNLOCK nor INV can be justified from it. That alone would call for a different predicate. The next subsection shows that it calls for different rules.

3.2 The printed rules are inconsistent

The derivations below use the dissertation's own points-to, $x \mapsto v$, and are checked against its own interpretation of assertions.

History can be invented. A model is a memory, not a record of how the memory came about. So a current cell is also a model of any stale value followed by that cell:

$$x \mapsto 7 * \mathbf{lock}_i \;\models\; x \mapsto 5 \lhd (x \mapsto 7 * \mathbf{lock}_i). \tag{1}$$

The stale cell is overridden by the later one, and nothing remains of it. This is the non-cancellativity that the dissertation notes in passing.

A skip that changes memory. Let J be $\mathbf{emp} \vee x \mapsto 7$. The following is a derivation in the printed calculus.

1. $\mathbf{emp} \vdash \{J * x \mapsto 7 * \mathbf{lock}_i\}\ \mathsf{skip}_i\ \{J * \mathbf{emp} * \mathbf{lock}_i\}$, by SKIP and CONS: J cannot be its second disjunct, since the cell would occur twice, and in the first disjunct the cell can be reassigned to J.
2. $J \vdash \{x \mapsto 7 * \mathbf{lock}_i\}\ \mathsf{skip}_i\ \{\mathbf{emp} * \mathbf{lock}_i\}$, by INV. The thread has given its cell to the invariant.
3. $J \vdash \{x \mapsto 5 \lhd (x \mapsto 7 * \mathbf{lock}_i)\}\ \mathsf{skip}_i\ \{x \mapsto 5 \lhd (\mathbf{emp} * \mathbf{lock}_i)\}$, by FRAME-TM.
4. $J \vdash \{x \mapsto 7 * \mathbf{lock}_i\}\ \mathsf{skip}_i\ \{x \mapsto 5 * \mathbf{lock}_i\}$, by CONS, using (1) on the left and the unit law for **emp** on the right.

The conclusion says that a thread holding the lock and a cell containing 7 may do nothing and then own that cell containing 5. In the state where processor 1 holds the lock and x contains 7, the precondition holds and the postcondition does not. (`printed_points_stale_value_unsound`; the mechanized statement parses INV's premise as $J * (P * \mathbf{lock}_i)$.)

A skip that drains another processor's buffer. Spatial conjunction takes the union of the completeness sets and does not require them to be disjoint, so an empty neighbour can copy a cell's marks: $x \mapsto 7 * \mathbf{emp} \models x \mapsto 7 * \mathbf{bar}_2$. With J the cell $x \mapsto 7$, SKIP, CONS and INV give $J \vdash \{\mathbf{emp} * \mathbf{lock}_1\}\ \mathsf{skip}_1\ \{\mathbf{bar}_2 * \mathbf{lock}_1\}$, and FRAME-TM with a pending write $R = y \rightsquigarrow_2 17$ gives

$$J \vdash \{R \lhd (\mathbf{emp} * \mathbf{lock}_1)\}\ \mathsf{skip}_1\ \{R \lhd (\mathbf{bar}_2 * \mathbf{lock}_1)\}.$$

When processor 1 holds the lock and processor 2's write to y is still buffered, the precondition holds. The postcondition places a barrier for processor 2 after its pending write, which that memory does not satisfy. (`printed_points_foreign_barrier_unsound`.)

The strong temporal frame rule does not escape. One might hope that FRAME-STM is safe, since its frame is also spatially separate. It is not, because $R \blacktriangleleft P$ abbreviates $(R * P) \wedge (R \lhd P)$ and the two conjuncts may be witnessed by different splits. Take $R = \mathbf{emp} \vee x \mapsto 5$. In the state above, the spatial conjunct of $R \blacktriangleleft (x \mapsto 7 * \mathbf{lock}_1)$ holds with R empty, and the temporal conjunct holds with R the stale cell. Applying FRAME-STM to step 2 yields a postcondition that the unchanged state cannot satisfy. (`printed_strong_frame_counterexample`; this derivation uses INV and FRAME-STM but not FRAME-TM.)

All three conclusions are also invalid for the printed safety predicate, but that is the lesser point. They are false under the dissertation's informal reading of a specification, and under any semantics for which a true specification describes what the program does.

3.3 Why no semantics could repair it

The first derivation is an instance of a general conflict.

Proposition 1 (`trilemma`)**.** *Suppose a logic of triples has an unrestricted temporal frame rule, the rule of consequence, and the entailment* $R \lhd \mathbf{emp} \models R$*. Suppose* $\{P\}\ c\ \{\mathbf{emp}\}$ *is provable. Then* $\{R \lhd P\}\ c\ \{R\}$ *is provable for every* R*. Hence if some execution of* c *starts in a state satisfying* $R \lhd P$ *and ends in a state that does not satisfy* R*, the logic is unsound.*

The proof is two rule applications. The content is in the last sentence. A triple $\{P\}\ c\ \{\mathbf{emp}\}$ with P owning a cell is a *transfer-out contract*: after c, the cell belongs to someone else. Whenever temporal conjunction lets later values override earlier ones, R may describe an older state of the same cell, as the store axiom's own postcondition does. Framing R across the transfer then hands the thread a stale copy of a cell it no longer owns.

So four things cannot all be kept: the unrestricted temporal frame rule, the right-unit law, transfer-out contracts ending in **emp**, and overlapping temporal history. A sound logic must give one up, and the choice determines its character:

- *No transfer out.* Cells may be taken from the invariant and never returned. This is sound and useless for a shared container.
- *No right unit.* Let **emp** record that a cell was given away. Framing survives, and every postcondition that mentions **emp** changes meaning.
- *Transfer leaves a mark.* Return an explicit tombstone assertion instead of **emp**. The algebra survives; every later specification carries the tombstones.
- *Restrict the frame.* Forbid temporal frames that overlap the cells being given away. Everything else is kept.

The first two were built and abandoned during this work, for the reasons given. The repaired calculus takes the fourth, and by Proposition 1 that restriction cannot be weakened: every frame that overlaps a released cell yields a false specification.

The second derivation has a different cause and a different repair. The all-processor completeness of points-to, and more generally the completeness marks of whatever the invariant holds, should not pass to a thread that acquires the lock. A thread can vouch only for its own buffer.

3.4 The dissertation's own example

Section 4.1 of the dissertation proves a small message-passing program correct, using an invariant that mentions two registers and the derived rule ATOMIC. That proof does not carry over to the repaired calculus: ATOMIC, SHARE and auxiliary-variable elimination are not available, and the concurrency rule now forbids either thread from modifying a variable free in the invariant. This shows that the proof does not port, not that its conclusion is false. [*pending:* the repaired derivation of the same program, and the recovery of its original conclusion]

4 What changed

Table 2 lists every difference between the printed calculus and the one proved sound. Each is marked as an error (the printed statement is false, with a checked counterexample), as forced by an error elsewhere, or as a choice among sound alternatives.

Three of these need a word of explanation.

Printed	Repaired	Kind
ASSIGN, LOAD, EX, CONC say nothing of $\mathrm{fv}(J)$	The assigned, loaded or bound variable, and every variable a parallel branch modifies, is absent from $\mathrm{fv}(J)$	error
List overriding by lookup; compatibility ignores blocked processors; pointwise implications	Ordered overriding; stable compatibility; implications quantify over futures (Table 1)	error
LOCK and UNLOCK exchange a token; INV moves cells	INV is removed; LOCK hands over J and UNLOCK takes it back	error
FRAME-TM for every specification	FRAME-TM only for specifications valid in *overlap mode*, which excludes UNLOCK	error
FRAME-STM	Omitted. Sound for one common split of the frame, unsound for $(R * P) \wedge (R \lhd P)$	error
SHARE	Omitted; adequacy states the top-level property directly [*pending:* restricted form]	forced
$e \mapsto e'$ with every processor complete	$e \rightarrowtail e'$, complete for some processor; load and store rules for it	choice
Any assertion as invariant	Invariants from a guarded, buffer-free fragment	choice
Any set of processors	At least two active processors	forced

Table 2: Differences between the printed and the repaired calculus.

Transferring lock rules. Section 3.1 observed that the printed predicate forgets the invariant at acquisition. The standard cure, from concurrent separation logic, is to move ownership with the lock: acquiring gives the thread the cells of J, and releasing requires it to give back cells satisfying J. INV then has nothing left to do. Removing it is also what blocks the three derivations of Section 3.2, all of which are proofs about skip: restricting the frame rules alone would not have been enough.

The cell. A thread that acquires the lock can vouch for its own buffer, which the machine required to be empty, and for no one else's. So the cells it receives carry its completeness mark only. The printed points-to demands every processor's mark and can therefore never be established by a command, only inherited from a precondition. The repaired calculus uses

$$e \rightarrowtail e' \;=_{df}\; \exists x : (e \rightsquigarrow_x e' \lhd \mathbf{bar}_x),$$

a flushed cell that is complete for whichever processor is credited with the write. This is the definition the dissertation considered first and rejected, on the grounds that $1 \rightsquigarrow_0 1 \lhd 2 \rightarrowtail 2$ would then have models in which the write to location 1 is still buffered. Those models are reachable, stable states of the machine. Because a model does not record who wrote a flushed cell, any marked processor may be credited, and two further axioms let any processor load from or store to such a cell. The printed points-to remains expressible and keeps its meaning.

Two processors. The lock is represented only by the set of blocked processors. With a single processor there is no one to block: a held lock is indistinguishable from a free one, $\mathbf{lock}_i$ is equivalent to **emp**, and $\mathsf{lock}_i \,;\, \mathsf{lock}_i$ runs to completion. A rule that hands over the invariant at acquisition would then hand it over twice. The assumption is not about finiteness, and the second processor need not execute anything.

Roads not taken. Two other repairs were built during this work and abandoned, and they are the first two exits of Section 3.3. Keeping the invariant's assertions in a separate compartment

of the judgment recovers the unrestricted FRAME-TM, but a cell can then never be returned to the invariant. Letting **emp** record cells that were given away permits the return, but $R \lhd \mathbf{emp}$ no longer entails R, and in that interpretation an ordinary later cell can silently erase an earlier, unrelated one.

5 The repaired model

The machine is unchanged. What changes is the predicate that says when a command is safe.

Resources. A thread's local state is a generalized memory ν with its *actual* completeness set. The printed predicate read every local state at $\Gamma = \mathbb{P}$, which is why it could not distinguish a barrier the thread had earned from one it had not. A mark $i \in \Gamma$ now means what the dissertation said informally: nothing of processor i's is pending before this resource. A flush of the resource's own oldest write by i earns the mark, as do fence_i, lock_i and unlock_i, whose guards require the whole of buffer i to be empty. Nothing else does.

Environments. The rest of the machine is described by a *pool* μ_J, holding the invariant's cells while the lock is free, and an external *frame* ν_F, standing for other threads and for any history that a frame rule has set aside. An environment for ν in mode m is a pool, a frame and a machine memory μ_0 with

$$\mu_0 \in \mu_J \mathbin{\widetilde{*}} (\nu_F \mathbin{\widehat{\#}} \nu)$$

such that either the lock is free in μ_0 and $(s, \mu_J, \mathbb{P}) \models J$, or the lock is held and μ_J is empty. In the held case the token may be in ν, in ν_F when another thread holds the lock, or split between them. The mode restricts the frame:

- in *overlap mode*, ov, any compatible frame is allowed, including one that mentions addresses of ν;
- in *disjoint mode*, dj, the frame and ν share no address.

Safety. $\mathit{safe}_{n+1}(c, s, \nu, J, Q)$ holds in mode m when, for every future $\nu_1 \leq \nu$:

1. if c is skip then $s, \nu_1 \models Q$; and
2. for every environment (μ_J, ν_F, μ_0) of ν_1, the configuration $c, (s, \mu_0)$ does not abort, and for every *program* step to $c', (s', \mu_1)$ there are ν' and μ'_J forming an environment of ν' with the *same* frame ν_F and memory μ_1, such that $\mathit{safe}_n(c', s', \nu', J, Q)$ and the step's bookkeeping conditions hold.

The bookkeeping conditions say that the pool changes only at lock and unlock; that unlock_i requires the whole token in ν_1; that the addresses of μ'_J and ν' lie within those of μ_J and ν_1; and that ν' gains no completeness mark beyond the one the step earns. A specification is valid in mode m, written $J \models^m \{P\}\ c\ \{Q\}$, when $s, \nu \models P$ implies safety at every n. Overlap-mode validity implies disjoint-mode validity.

Four features distinguish this from the printed predicate.

- *The frame is fixed within a step and arbitrary between steps.* The printed predicate let the frame take silent steps of its own choosing during a program step. That permits a local residual to claim a pending write that belonged to the frame, and the frame lemma fails (`overlap_frame_shape_counterexample`). Re-quantifying the frame at every step is what lets another thread, including one holding the lock, act in between.

- *Flushes are handled by closure.* A silent step is attributed to the frame or to the local resource with the other fixed (`pooled_resource_flush_attribution`), and safety is closed under futures of the local resource by its outermost quantifier.
- *The mark budget.* Without it, a proof could choose a successor that claims another processor's barrier.
- *The modes.* UNLOCK is valid only in disjoint mode: the released cells become the pool, and a frame holding a stale copy of one would collide with it. That is Proposition 1 seen from the semantics.

Three lemmas carry the proofs. With the frame fixed, the local residual's buffers are determined by the machine's: ordered overriding places the frame's writes to an address before the local ones, so there is one way to subtract the frame (`ordered_residual_unique`). Every machine flush belongs to exactly the frame or the local resource. And an invariant from the guarded fragment (Section 6) is buffer-free and holds of a memory regardless of which processors are marked complete (`invariant_fragment_remark`), which is what allows the pool to be read at full completeness and handed over with the acquirer's mark alone.

6 The repaired calculus

The judgment is $J \vdash^m \{P\}\ c\ \{Q\}$, where m is ov or dj. It has 21 rules. The rules SKIP, ASSUME, ASSERT, FENCE, STORE, SEQ, CONS, DISJ, CHOICE, LOOP and FRAME-SP are those of Figure 1 at either mode. Figure 2 shows the rest; Appendix A has the complete system.

$$\textsc{lock}\ \frac{}{J \vdash^m \{\mathbf{emp}\}\ \mathsf{lock}_i\ \{(\mathbf{lock}_i * J) \lhd \mathbf{bar}_i\}} \qquad \textsc{unlock}\ \frac{A \lhd \mathbf{bar}_i \models J * B}{J \vdash^{\mathsf{dj}} \{\mathbf{lock}_i * A\}\ \mathsf{unlock}_i\ \{B\}}$$

$$\textsc{frame-tm}\ \frac{J \vdash^{\mathsf{ov}} \{P\}\ c\ \{Q\} \qquad \mathrm{mod}(c) \cap \mathrm{fv}(R) = \emptyset}{J \vdash^{\mathsf{ov}} \{R \lhd P\}\ c\ \{R \lhd Q\}} \qquad \textsc{weaken}\ \frac{J \vdash^{\mathsf{ov}} \{P\}\ c\ \{Q\}}{J \vdash^{\mathsf{dj}} \{P\}\ c\ \{Q\}}$$

$$\textsc{load-any}\ \frac{x \notin \mathrm{fv}(J, e, e', P)}{J \vdash^m \{e \rightarrowtail e' \blacktriangleleft P\}\ x := [e]_i\ \{(e \rightarrowtail e' \blacktriangleleft P) \wedge x = e'\}}$$

$$\textsc{store-any}\ \frac{}{J \vdash^m \{e \rightarrowtail e'' \blacktriangleleft P\}\ [e] := e'_i\ \{(e \rightarrowtail e'' \blacktriangleleft P) \lhd e \rightsquigarrow_i e'\}} \qquad \textsc{assign}\ \frac{x \notin \mathrm{fv}(J)}{J \vdash^m \{P\,[e/x]\}\ x := e_i\ \{P\}}$$

$$\textsc{ex}\ \frac{J \vdash^m \{P\}\ c\ \{Q\} \qquad x \notin \mathrm{fv}(J, c, Q)}{J \vdash^m \{\exists x : P\}\ c\ \{Q\}}$$

Figure 2: Rules of the repaired calculus that differ from Figure 1. LOAD is the printed axiom with $x \notin \mathrm{fv}(J, e, i, e', P)$, and CONC is the printed rule at a common mode with $\mathrm{mod}(c_k) \cap \mathrm{fv}(J) = \emptyset$ added.

Lock and unlock. LOCK delivers the invariant together with the acquirer's barrier. UNLOCK takes the whole of the thread's state A, notes that releasing drains the thread's buffer, and asks for

a split into something satisfying J and a remainder B. Because A is arbitrary, nothing needs to be framed around a release: whatever the thread wants to keep goes into A and comes out in B.

Invariants. The soundness theorem requires J to come from the fragment

$$H \ ::= \ \mathbf{emp} \mid e \rightarrowtail e' \mid H * H \mid H \vee H \mid \exists x : H \mid b \wedge H.$$

A pure formula may appear only alongside a heap formula. A bare **true** is not buffer-free: it holds of a memory with pending writes, and an invariant containing it could swallow them at release.

Modes in practice. A specification for a command that never releases can be derived in overlap mode and framed temporally. Once a release is sequenced in, the whole is in disjoint mode and can only be framed spatially. Overlap-mode pieces are weakened before sequencing. At top level the distinction disappears (Section 7).

What became of the derived rules. The dissertation lists a dozen derived and alternative rules. In the repaired calculus LOAD-P is derived exactly as displayed, with the printed points-to; FENCE-G is derived as displayed; LOCK-G, LOCK-B and the LOCKED and ATOMIC wrappers are derived in a form that carries the transferred invariant; and the unlock variants are derived for the empty invariant, the general case being UNLOCK itself. The "daring" rules for unlocked access to shared state remain unavailable. Appendix C has the table.

7 Soundness and adequacy

Theorem 1 (Soundness, `owned_calculus_sound`)**.** *Let J be in the invariant fragment and let there be two distinct active processors. If $J \vdash^m \{P\}\ c\ \{Q\}$ is derivable then $J \models^m \{P\}\ c\ \{Q\}$.*

The proof is by induction on the derivation, one semantic lemma per rule. No rule has a premise about safety or validity. CONS and UNLOCK have semantic entailments as premises. [*pending:* assertion proof system]

Theorem 2 (Adequacy, `owned_derivation_adequacy`)**.** *Under the same hypotheses, let $J \vdash^m \{P\}\ c\ \{Q\}$ be derivable. Take a machine memory with the lock free that splits spatially into a pool satisfying J and a resource satisfying P. Then no finite execution of c from that memory aborts. If an execution terminates, its final memory splits into a pool and a resource satisfying Q, where either the lock is free and the pool satisfies J, or the resource holds the lock and the pool is empty.*

Adequacy instantiates the external frame as empty, which satisfies both modes, so a client never discharges a mode condition. This theorem does the work that SHARE did in the dissertation: it turns a specification under J into a statement about the whole memory. It also covers the case that makes SHARE unsound next to the transferring LOCK: a command that ends with the lock held has the invariant in its own hands, and SHARE would count it twice (`owned_printed_share_lock_counterexample`).

For a specification from **emp** to **emp** there is a sharper statement (`owned_emp_derivation_restores`): every completed execution ends with the lock free and the final memory is exactly an invariant pool together with whatever the environment's frame has become.

These are partial-correctness results. They say nothing of termination, fairness or deadlock, and an execution stopped inside a critical section need not satisfy the invariant.

What the semantic rules need. The standing hypotheses are used unevenly, and less than one might expect. The frame rules, the concurrency rule and the heap axioms need neither two processors nor, except for FRAME-TM, the fragment. The concurrency rule is proved for both modes and for nested parallel composition (`owned_parallel_valid`); when one thread finishes, its postcondition is carried through the rest of the other thread's execution by the spatial frame rule.

8 How proofs proceed

The dissertation gave few examples. This section gives three, in increasing size. All are derivations in the calculus of Section 6, checked by Rocq.

8.1 A store, with and without a fence

For any invariant and either mode,

$$J \vdash^{m} \{x \rightarrowtail u\}\ [x] := v_i\,;\mathsf{fence}_i\ \{x \rightarrowtail v\}$$

is derivable (`owned_derive_store_fence_cell`): STORE-ANY gives $x \rightarrowtail u \lhd x \rightsquigarrow_i v$, FRAME-TM carries that history across fence_i, and a drained history collapses to its last write. This is the store rule of ordinary separation logic. The same specification without the fence is *invalid* (`owned_plain_store_cell_invalid`): the store may terminate with its write still buffered, and a cell assertion has no buffered writes. The logic is the difference between these two.

8.2 Reading a buffered write

After $[x] := v_i$ the thread holds $R \lhd x \rightsquigarrow_i v$, where R describes the older state of x. The load axiom wants the loaded write to be spatially separate from the rest, and an old cell cannot be separated spatially from a newer write to the same address. The temporal frame rule is what resolves this. The newest write alone satisfies the axiom's precondition, with **emp** for P; the older history is a temporal frame:

$$\frac{J \vdash^{\mathsf{ov}} \{x \rightsquigarrow_i v\}\ r := [x]_i\ \{x \rightsquigarrow_i v \wedge r = v\}}{J \vdash^{\mathsf{ov}} \{R \lhd x \rightsquigarrow_i v\}\ r := [x]_i\ \{R \lhd (x \rightsquigarrow_i v \wedge r = v)\}}\ \text{FRAME-TM}$$

(`owned_derive_load_after_history`). Store-to-load forwarding, the behaviour most characteristic of the machine, is thus an instance of the rule that most distinguishes the logic, and the reason it was worth keeping.

8.3 A shared container

One cell x is shared through a flag f. With $X = \exists v : x \rightarrowtail v$, the invariant is

$$J \;=\; f \rightarrowtail 0 \;\vee\; (f \rightarrowtail 1 * X).$$

A thread on processor 1 takes the item, writes to it privately, and returns it:

$$\begin{aligned}&\mathsf{lock}_1;\ r := [f]_1\,;\ \mathsf{assume}(r = 1)_1;\ [f] := 0_1;\ \mathsf{unlock}_1;\\ &[x] := 9_1;\\ &\mathsf{lock}_1;\ r := [f]_1\,;\ \mathsf{assume}(r = 0)_1;\ [f] := 1_1;\ \mathsf{unlock}_1\,.\end{aligned}$$

Meanwhile processor 2 has an unrelated write $R = y \rightsquigarrow_2 17$ buffered. The derived specification is $J \vdash^{\mathsf{dj}} \{R\}\ c\ \{R\}$ (`owned_container_derived`). Writing $H(a, v)$ for $(\exists u : a \mapsto u) \lhd a \rightsquigarrow_1 v$, the proof passes through these assertions, each with R as a spatial frame:

After	The thread holds
first load and assume	$\mathbf{lock}_1 * (f \mapsto 1 * X)$
$[f] := 0$	$\mathbf{lock}_1 * (H(f, 0) * X)$
first unlock	X
$[x] := 9$	$H(x, 9)$
second lock	$H(x, 9) \lhd ((\mathbf{lock}_1 * J) \lhd \mathbf{bar}_1)$
second load and assume	$\mathbf{lock}_1 * (f \mapsto 0 * x \mapsto 9)$
$[f] := 1$	$\mathbf{lock}_1 * (H(f, 1) * x \mapsto 9)$
second unlock	$\mathbf{emp}$

Three things are visible in it.

The foreign write costs one rule application. R concerns an address the thread never touches, so a single FRAME-SP around the entire program carries it, across both releases.

The temporal frame rule carries the private write across the second acquisition. $H(x, 9)$ is overlapping history, possibly still buffered. FRAME-TM places it before the acquired state, and the barrier that LOCK delivers then drains it, leaving $x \mapsto 9$.

Releases use entailment, not framing. The two UNLOCK premises are $(H(f, 0) * X) \lhd \mathbf{bar}_1 \models J * X$ and $(H(f, 1) * x \mapsto 9) \lhd \mathbf{bar}_1 \models J * \mathbf{emp}$. The first keeps the item and the second donates it.

The example is not vacuous. From a memory with the lock free, $f = 1$, $x = 7$ and processor 2's write pending, there is a complete 24-step execution in which that write is pending in every state (`owned_container_foreign_pending_throughout`), and by adequacy every execution from that memory is safe and, if it completes, restores the invariant.

A second version replaces each `assume()` by a loop that releases the lock and retries when the flag is wrong (`owned_retrying_container_derived`). Its return loop has a branch for "the container is already full", and that branch is derived from **false**: the thread holds x, so the pool cannot.

The cost. The container's derivation applies CONS ten times and needs sixteen entailments that are not identities, proved here semantically. Most are rearrangements, a few are the laws about barriers and histories used above, and one, at the second acquisition, takes fifty lines. [*pending:* the same derivation with syntactic entailments; the Section 4.1 example; named locks and a producer/consumer client]

9 The mechanization

The development is about 25,000 lines of Rocq in 148 files, using only the standard library. It declares no axioms and admits no proofs. An audit file prints the assumptions of every principal result, 947 at the time of writing, and the build checks the compiled proof objects independently. The proofs need neither extensionality nor choice nor a finite set of processors. [*pending:* final figures]

The transcription of the printed definitions is kept separate from every repair, so that each claim in Section 3 is a theorem about the dissertation's own definitions and not about an approximation of them.

The work was done with AI assistance, in a loop that turned out to matter. One model developed the proofs; a second reviewed them independently and proposed the design; each rule

or lemma was stated before it was proved, and where a bounded search was cheap it was run first. Twice the search or the proof attempt refuted a confident argument on paper: a frame lemma that let the frame evolve freely, and the soundness of FRAME-STM. Both now appear above as counterexamples. I do not think either would have been caught by reading. [*pending:* Ian to confirm how this paragraph should read]

10 Later work

[*pending:* a short, sourced comparison with program logics for weak memory published since 2012; to be written after a literature pass]

11 What remains open

Unlocked access to shared state. The dissertation opens with Dekker's algorithm, made correct on this machine by a fence after each store. Each thread reads a flag that the other writes, with no lock. The calculus cannot express that: a thread may touch a cell only if it owns it or holds the lock. The dissertation's "daring" rules were meant for this and were left as future work; they still are.

Realistic locks. The one lock is global and blocks every other processor. Programmer-level locks can be encoded as flag cells under the invariant, as the container does, but each release must itself take the global lock. A spinlock released by a bare store is another instance of unlocked access.

A strong temporal connective. The frame property holds for a single split that is both spatial and temporal (`owned_safe_common_frame`). Making that a connective of the assertion language, in place of the conjunction of two, would restore FRAME-STM. Whether its other laws would then hold is not known.

The rest of the dissertation. Allocation, recursive predicates, permissions and the other topics of Chapter 5 are untouched, as is the sequential logic of Chapter 3.

References

[1] Ian Wehrman. *Weak-Memory Local Reasoning.* PhD dissertation, The University of Texas at Austin, December 2012.

A The repaired calculus in full

Figure 3 gives every rule of the calculus proved sound in Section 7.

SKIP
$$\frac{}{J \vdash^m \{P\}\ \mathsf{skip}_i\ \{P\}}$$

ASSUME
$$\frac{}{J \vdash^m \{\neg b \vee P\}\ \mathsf{assume}(b)_i\ \{P\}}$$

ASSERT
$$\frac{}{J \vdash^m \{b \wedge P\}\ \mathsf{assert}(b)_i\ \{P\}}$$

ASSIGN
$$\frac{x \notin \mathrm{fv}(J)}{J \vdash^m \{P\,[e/x]\}\ x := e_i\ \{P\}}$$

FENCE
$$\frac{}{J \vdash^m \{\mathbf{emp}\}\ \mathsf{fence}_i\ \{\mathbf{bar}_i\}}$$

LOCK
$$\frac{}{J \vdash^m \{\mathbf{emp}\}\ \mathsf{lock}_i\ \{(\mathbf{lock}_i * J) \lhd \mathbf{bar}_i\}}$$

UNLOCK
$$\frac{A \lhd \mathbf{bar}_i \models J * B}{J \vdash^{\mathsf{dj}} \{\mathbf{lock}_i * A\}\ \mathsf{unlock}_i\ \{B\}}$$

LOAD
$$\frac{x \notin \mathrm{fv}(J, e, i, e', P)}{J \vdash^m \{e \rightsquigarrow_i e' \blacktriangleleft P\}\ x := [e]_i\ \{(e \rightsquigarrow_i e' \blacktriangleleft P) \wedge x = e'\}}$$

STORE
$$\frac{}{J \vdash^m \{e \rightsquigarrow_i e'' \blacktriangleleft P\}\ [e] := e'_i\ \{(e \rightsquigarrow_i e'' \blacktriangleleft P) \lhd e \rightsquigarrow_i e'\}}$$

LOAD-ANY
$$\frac{x \notin \mathrm{fv}(J, e, e', P)}{J \vdash^m \{e \rightarrowtail e' \blacktriangleleft P\}\ x := [e]_i\ \{(e \rightarrowtail e' \blacktriangleleft P) \wedge x = e'\}}$$

STORE-ANY
$$\frac{}{J \vdash^m \{e \rightarrowtail e'' \blacktriangleleft P\}\ [e] := e'_i\ \{(e \rightarrowtail e'' \blacktriangleleft P) \lhd e \rightsquigarrow_i e'\}}$$

FRAME-SP
$$\frac{J \vdash^m \{P\}\ c\ \{Q\} \qquad \mathrm{mod}(c) \cap \mathrm{fv}(R) = \emptyset}{J \vdash^m \{R * P\}\ c\ \{R * Q\}}$$

FRAME-TM
$$\frac{J \vdash^{\mathsf{ov}} \{P\}\ c\ \{Q\} \qquad \mathrm{mod}(c) \cap \mathrm{fv}(R) = \emptyset}{J \vdash^{\mathsf{ov}} \{R \lhd P\}\ c\ \{R \lhd Q\}}$$

WEAKEN
$$\frac{J \vdash^{\mathsf{ov}} \{P\}\ c\ \{Q\}}{J \vdash^{\mathsf{dj}} \{P\}\ c\ \{Q\}}$$

CONS
$$\frac{P \models P' \qquad J \vdash^m \{P'\}\ c\ \{Q'\} \qquad Q' \models Q}{J \vdash^m \{P\}\ c\ \{Q\}}$$

SEQ
$$\frac{J \vdash^m \{P\}\ c\ \{R\} \qquad J \vdash^m \{R\}\ c'\ \{Q\}}{J \vdash^m \{P\}\ c\,;c'\ \{Q\}}$$

CHOICE
$$\frac{J \vdash^m \{P\}\ c\ \{Q\} \qquad J \vdash^m \{P\}\ c'\ \{Q\}}{J \vdash^m \{P\}\ c + c'\ \{Q\}}$$

LOOP
$$\frac{J \vdash^m \{P\}\ c\ \{P\}}{J \vdash^m \{P\}\ c^*\ \{P\}}$$

DISJ
$$\frac{J \vdash^m \{P\}\ c\ \{Q\} \qquad J \vdash^m \{P'\}\ c\ \{Q\}}{J \vdash^m \{P \vee P'\}\ c\ \{Q\}}$$

EX
$$\frac{J \vdash^m \{P\}\ c\ \{Q\} \qquad x \notin \mathrm{fv}(J, c, Q)}{J \vdash^m \{\exists x : P\}\ c\ \{Q\}}$$

CONC
$$\frac{J \vdash^m \{P_k\}\ c_k\ \{Q_k\} \qquad \mathrm{mod}(c_{1-k}) \cap \mathrm{fv}(J, P_k, c_k, Q_k) = \emptyset \quad (k = 0, 1)}{J \vdash^m \{P_0 * P_1\}\ c_0 \,\|\, c_1\ \{Q_0 * Q_1\}}$$

Figure 3: The 21 rules of the repaired calculus (`owned_derivation`). In LOCK and UNLOCK the processor is a constant.

B The dissertation's assertion laws

Table 3 classifies every law of the dissertation's Figures 4.5 and 4.6 in the final interpretation. A proved law holds for all assertions, implications included, and all processor domains. Each refuted law is also refuted in the dissertation's original interpretation.

Law ($\bullet$ ranges over $*$, $\lhd$, $\blacktriangleleft$)	Status	Note
$P \bullet \mathbf{emp} \equiv P \equiv \mathbf{emp} \bullet P$	proved	all three
$(P \bullet Q) \bullet R \equiv P \bullet (Q \bullet R)$	proved for $*$, $\lhd$	refuted for $\blacktriangleleft$, both directions
$P * Q \equiv Q * P$	proved	
$P \bullet \mathbf{lock}_e \equiv \mathbf{lock}_e \bullet P$	proved	indeed $P \lhd \mathbf{lock}_e \equiv P * \mathbf{lock}_e$
$\mathbf{lock}_e \bullet \mathbf{lock}_f \equiv \mathbf{false}$	qualified	needs a processor blocked by both
$\mathbf{bar}_e \bullet \mathbf{bar}_e \equiv \mathbf{bar}_e$	proved	
$(P \bullet Q) \lhd \mathbf{bar}_e \equiv (P \lhd \mathbf{bar}_e) \bullet (Q \lhd \mathbf{bar}_e)$	proved	all three
$P \blacktriangleleft \mathbf{bar}_e \equiv P \lhd \mathbf{bar}_e$	proved	
$\mathbf{lock}_e \lhd \mathbf{bar}_e \equiv \mathbf{lock}_e * \mathbf{bar}_e$	proved	
$e \rightsquigarrow_{e'} e'' * e \rightsquigarrow_{f'} f'' \equiv \mathbf{false}$	proved	
$e \rightsquigarrow_i e' \lhd e \rightsquigarrow_i f \lhd \mathbf{bar}_i \equiv e \rightsquigarrow_i f \lhd \mathbf{bar}_i$	proved	
$\mathbf{bar}_e \models \mathbf{emp}$; monotonicity of $\bullet$	proved	
$e \mapsto f \models e \rightsquigarrow_{e'} f$	qualified	e' must denote an active processor
$P \blacktriangleleft Q \models P * Q$ and $P \blacktriangleleft Q \models P \lhd Q$	proved	
Small exchange, $\blacktriangleleft$ outermost	refuted	under either reading of its premise
Small exchange between $*$ and $\lhd$	vacuous	its premise forces an empty processor set
Full exchange for $*$ and $\blacktriangleleft$	refuted	

Table 3: The printed assertion laws, classified.

The pattern is simple to state. Every law that rebrackets or exchanges through $\blacktriangleleft$ fails, for the reason FRAME-STM fails: the two conjuncts of $P \blacktriangleleft Q$ choose their splits independently. Everything else survives, two laws with a side condition. The unit laws needed the most care, since composing with an empty resource can add completeness marks, and the proof must show that no assertion, however many implications it contains, can tell.

C The dissertation's derived rules

Printed rule	In the repaired calculus
LOAD-P	Derived as displayed, with the printed points-to.
STORE-P	The display has a load command and a strong temporal suffix, which no memory satisfies; with a store command and a temporal suffix, as in STORE, it is derived.
Strengthened LOCK, UNLOCK	Subsumed: LOCK already delivers $\mathbf{bar}_i$, and UNLOCK's premise already assumes it.
FENCE-G	Derived as displayed.
LOCK-G, LOCK-B	Derived with the transferred invariant made explicit.
UNLOCK-G, UNLOCK-B	Derived for the empty invariant; in general the thread cannot promise to keep all of P while donating part of it.
FENCE-B	The display has lock where fence is meant; with fence it is derived.
LOCKED, ATOMIC	Derived as a wrapper whose body is proved under the same J. Importing a body proved under **emp** is not recovered.
Daring load and store	Not available (Section 11).

Table 4: Status of the rules in the dissertation's "Derived and Alternative Axioms and Inference Rules".